

Weijia Zhang

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EDUCATION

Yale University

M.S. in Computer Science (Thesis Track with Full Scholarship)

University of Illinois Urbana-Champaign (UIUC)

B.S. in Computer Science and Mathematics (C.W. Gear Outstanding Undergraduate Award, 1 of 2/yr)

Sep 2026 - May 2028

GPA: 4.0/4.0

Aug 2022 - May 2026

GPA: 3.7/4.0 (2025 Dean's List)

WORK EXPERIENCE

TikTok

Research Scientist Intern, RSI/Self-Evolving Agents

Jun. 2026 – Present

- As project lead, designed and shipped TikTok's agent self-evolution framework from 0 to 1 with a closed *Solver-Reflector-Evolver* loop: the Solver executes tasks, the Reflector attributes failures, and the Evolver iterates reusable skills; lifted F1 from **62.7% to 81.3%** on unoriginal-video detection (9,600 real short videos) and from **76.5% to 87.3%** on reposting-account detection (200 accounts)
- Designed a transferable **Online Failure Discovery**: Reflector mines failure patterns directly from scenarios and Solver agent traces, builds and continuously updates a two-level "scenario → agent failure mode" taxonomy, and distills high-value error clusters into reusable skills; the same framework therefore self-evolves on scenarios such as unoriginal-video, reposting-account detection, without any human-predefined error types
- Trained a **video-understanding model**: built an account-level SFT dataset of 74K accounts and 1.33M videos, and ran distributed full-parameter SFT of Qwen3-VL 2B/4B/8B on 32 H100s with DeepSpeed, plus cross-domain evaluation and threshold calibration over 53 checkpoints; the final 4B model cut false positives by **42.8%** (318 → 182) against the 8B baseline at ~70% recall

Microsoft

Research Intern, Large Language Models

July 2025 – Sep 2025

- Built the **Excel Coding Agent** data engine producing execution-verified SFT/RL data for Excel Copilot's code agent: mined 20,000+ real-world workbooks, back-translated user queries from spreadsheet context, sampled multi-turn Office.js rollouts (write → execute → repair), and used SheetEngine final-state assertions as a programmatic verifier to filter incorrect and reward-hacking solutions
- Post-trained the coding agent on this corpus with rejection-sampling SFT and RL over same-task pass/fail rollouts under execution-based reward, improving Office Scripts **pass@1 by 15%** against the production baseline on a held-out set

Reborn Network

AI Agent Engineer

May 2023 – July 2023

- Built an **embodied role-playing agent** in a Unity VR environment, closing a real-time perception-dialogue-action loop across text, voice, and full-body VR actions at sub-second end-to-end latency (< 1s)
- Introduced RAG/vector database and dual-level (episodic + semantic) memory, improving **cross-session recall accuracy from 38% to 61%** on an internal multi-session dialogue eval

PUBLICATIONS (200+ CITATIONS)

Weijia Zhang et al. "CUADebug: Diagnosing and Repairing Computer-Use Agent Failures" *EMNLP 2026*.

Weijia Zhang et al. "Every Act Has Its Price: Compressed Moral Composition in Frontier LLMs." *EMNLP 2026*.

Weijia Zhang et al. "How Much Vision Does Multimodal Reasoning Need?" *Under review at NeurIPS 2026*.

Weijia Zhang*, Zijia Liu*, Haoru Li*, et al. "SeeingEye: Agentic Information Flow Unlocks Multimodal Reasoning in Text-only LLMs." *arXiv:2510.25092; under review at AAAI 2026*.

Weijia Zhang* and Gordon Dai* et al. "Artificial Leviathan: Exploring Social Evolution of LLM Agents Through the Lens of Hobbesian Social Contract Theory." *AI Behavioral Science Workshop, ACM SIGKDD 2024*.

Miaomiao Li et al. (including **Weijia Zhang**). "Understanding and Mitigating the Bias Inheritance in LLM-based Data Augmentation on Downstream Tasks." *ACL 2026. Oral*.

Kunlun Zhu et al. (including **Weijia Zhang**). "Where LLM Agents Fail and How They Can Learn From Failures." *arXiv:2509.25370. Failure Modes in Agentic AI, ICML 2026*

PROJECT EXPERIENCE

OpenManus & OpenManus-RL (GitHub 70k Stars) | Core Author | [GitHub](#)

Jan 2025 – Jul 2025

- Collaborated with the official OpenManus team as a core contributor to OpenManus-RL, a **verl**-based agent RL post-training framework; the ecosystem has accumulated 70,000+ GitHub stars
- Connected SFT, RL(PPO/GRPO), interactive environments, trajectory rollout into one unified pipeline to train react agent and tool-use on long-horizon tasks such as GAIA, WebShop, and ALFWorld
- Unified multi-turn environment interaction, tool actions, validation, reward under a single ReAct trajectory protocol, and released a dataset of **~49K high-quality agent trajectories** spanning OS, Web, Database, Knowledge Graph, Household, and E-commerce scenarios

AgentDebugX (Github 100 Stars) | Project Lead | [GitHub](#)

Aug 2026 – Present

- As co-lead, designed and implemented **AgentDebugX**, the **first comprehensive debugging framework for long-horizon agent trajectories**; its *Detect-Attribute-Recover-Rerun* loop turns LLM agent execution traces into auditable root-cause diagnoses, repair suggestions, and verifiable reruns across multi-agent, tool-using, and computer-use agents
- Designed the core diagnostic agent **DeepDebug**, which localizes root causes over multiple rounds via global trajectory investigation; on the 184 Who&When trajectories it raises agent-and-step attribution accuracy from **21.7% to 28.8%**, and on GAIA it repairs **13/73** failed tasks in a single round, lifting overall accuracy from **55.8% to 63.6%**
- Built an open-source infrastructure spanning a Python library, CLI, Computer-use, agentic skill, and Error Hub, providing unified trajectory ingestion, failure reuse, and regression validation supporting LangGraph, CrewAI, OpenAI Agents SDK, OpenTelemetry, and OSWorld

SKILLS

Programming Languages: Python, C/C++, C#

Frameworks & Libraries: VERL, vLLM, LangGraph, LangChain, PyTorch, TensorFlow

AI Focus: RL/Post-training, LLM Agents, Multimodal Reasoning, Self-Evolving Agents, SFT, RAG, Evaluation